

# Root Cause Analysis (RCA) Report

Blameless Postmortem — Cloud / DevOps Incident (Technical)

## 1. Incident Metadata

|                               |                                          |                      |                     |
|-------------------------------|------------------------------------------|----------------------|---------------------|
| Incident ID                   | INC-2026-0142                            | Severity (SEV1–SEV4) | SEV-2               |
| Date of Incident              | 12 Aug 2026, 03:10 AM IST                | Date RCA Completed   | 13 Aug 2026         |
| Incident Commander / Reporter | On-Call SRE                              | RCA Owner / Author   | Cloud Platform Team |
| Systems / Services Affected   | prod-vnet, resource-sync-service (Azure) | Related Ticket(s)    | JIRA-4521           |
| Detection Time                | 03:12 AM IST                             | Resolution Time      | 03:31 AM IST        |
| Total Impact Duration         | 19 minutes                               | Environment          | Production          |
| Customer-Facing? (Y/N)        | Yes                                      | Status               | Final               |

## 2. Incident Summary

- Nightly Terraform drift-remediation job auto-applied a plan on prod-vnet
- Subnet CIDR had been manually edited in Azure Portal, causing state drift
- Terraform destroyed and recreated the subnet instead of an in-place update
- Recreation detached 3 NSGs, breaking resource-sync-service connectivity
- 2 customer subscriptions affected for 19 minutes

## 3. Impact

- Services affected: resource-sync-service (multi-cloud inventory sync)
- Customers affected: 2 enterprise tenants (~340 end users)
- Duration: 19 minutes (03:10 – 03:29 AM IST)
- Business impact: SLA breach on both tenants; no data loss; no revenue impact

## 4. Timeline of Events

| Time (IST) | Event                                                       |
|------------|-------------------------------------------------------------|
| 03:10 AM   | Terraform drift-remediation job starts (GitLab CI cron)     |
| 03:11 AM   | terraform apply destroys and recreates prod-subnet-payments |
| 03:12 AM   | PagerDuty alert: resource-sync-service health check failing |
| 03:14 AM   | On-call SRE acknowledges, begins investigation              |
| 03:19 AM   | Root cause found: NSGs detached from recreated subnet       |
| 03:24 AM   | NSGs manually re-associated via Azure CLI                   |
| 03:29 AM   | Health checks pass; connectivity restored                   |
| 03:31 AM   | Incident resolved                                           |

## 5. Detection

- Detected via automated PagerDuty alert, not a manual report
- Time to detection: ~2 minutes
- Detection speed was adequate; no improvement needed here

## 6. Investigation & Troubleshooting Commands

Step 1 — Check pod / service health:

```
$ kubectl get pods -n prod -l app=resource-sync-service
→ check pod status

$ kubectl logs -n prod deploy/resource-sync-service --tail=100
→ check for connection errors
```

Step 2 — Inspect Terraform state for the subnet:

```
$ terraform state list | grep subnet
→ list subnet resources in state

$ az network vnet subnet show -g prod-rg --vnet-name prod-vnet -n prod-subnet-payments --query id
```

Step 3 — Confirm NSG detachment:

```
$ az network vnet subnet show -g prod-rg --vnet-name prod-vnet -n prod-subnet-payments --query
networkSecurityGroup
→ returns null → NSG detached

$ az network nsg list -g prod-rg -o table
→ list NSGs available to re-attach
```

Step 4 — Re-attach NSG:

```
$ az network vnet subnet update -g prod-rg --vnet-name prod-vnet -n prod-subnet-payments --network-
security-group prod-nsg-payments
```

Step 5 — Verify recovery:

```
$ kubectl rollout restart deploy/resource-sync-service -n prod
→ force pods to reconnect

$ kubectl get pods -n prod -l app=resource-sync-service -w
→ watch pods return healthy

$ curl -s -o /dev/null -w "%{http_code}" https://resource-sync.internal/healthz
→ 200 = confirmed recovery
```

## 7. Root Cause Analysis (5 Whys)

|                    |                                                                                                                                               |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| Why 1              | resource-sync-service lost connectivity because 3 NSGs were detached from prod-subnet-payments.                                               |
| Why 2              | The NSGs detached because Terraform destroyed and recreated the subnet instead of updating it in place.                                       |
| Why 3              | Terraform chose destroy-and-recreate because the subnet's live CIDR no longer matched the CIDR in Terraform state — a drift.                  |
| Why 4              | The drift existed because an engineer manually changed the CIDR in the Azure Portal two weeks earlier, without updating Terraform.            |
| Why 5 (Root Cause) | No policy gate blocked manual out-of-band changes, and no drift-detection alerting existed before the nightly job applied the fix unattended. |

## 8. Contributing Factors

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- No approval gate on the nightly automated drift-remediation apply
- No pre-apply check flagging 'destroy' actions on prod-tagged resources
- No Slack notification before an unattended production apply ran
- Manual portal change was never flagged by any drift-detection alert

## 9. Resolution

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- Re-attached the 3 detached NSGs via Azure CLI (Section 6, Steps 3–4)
- Verified with synthetic traffic tests and a direct health-check curl
- No data lost — recreated subnet had no stateful resources attached

Post-fix validation:

```
$ terraform plan -target=azurerm_subnet.payments  
→ confirms no further drift
```

## 10. Action Items

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| Action Item                                                       | Owner         | Priority |
|-------------------------------------------------------------------|---------------|----------|
| Require approval for any TF plan with 'destroy' on prod resources | Platform Team | P0       |
| Add daily drift-detection alerting                                | On-Call SRE   | P1       |
| Block manual portal edits on TF-managed resources (Azure Policy)  | Platform Team | P1       |

## 11. Sign-off

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| RCA Owner           | Reviewed By   | Date Approved |
|---------------------|---------------|---------------|
| Cloud Platform Team | Platform Lead | 14 Aug 2026   |